



COMPETENCY STANDARD

FOR

CAD-CAM Design and Programming (Mechanical)

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for CAD-CAM Design and Programming (Mechanical) serve as a foundational document for structuring curricula, developing teaching and learning materials, and designing assessment tools. It supports training initiatives aligned with industry needs, ensuring that individuals who complete these standards through proper training and assessment are adequately equipped for relevant roles in the Light Engineering sector.

This document was initially developed under **the Skills for Employment Investment Program (SEIP)** and has been updated and refined for use in training programs under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. These updates aim to address evolving industry demands and enhance workforce capabilities. The ownership of this document resides with the Finance Division of the Ministry of Finance, People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the

workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CAD-01-G	Apply Occupational Health and Safety (OHS) Practices in Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-CAD-02-G	Communicate in English in the Workplace	1. Read and understand workplace documents in English 2. Prepare simple workplace documents in English 3. Listen and comprehend to English conversations 4. Perform conversations in English	5
SICIP-LE-CAD-03-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	5
Total Hour			20

Sector Specific Competencies (28 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CAD-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	08
SICIP-LE-CAD-02-S	Work with Hand Tools and Power Tools	1. Inspect hand tools and power tools for usability 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	08
SICIP-LE-CAD-03-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate work measurements 3. Execute simple work calculations 4. Clean and maintain measuring instruments	12
Total Hour			28

OCCUPATION SPECIFIC COMPETENCIES (290 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-CAD-01-O	Understand mechanical drawing	1. Interpret mechanical drawing 2. Create mechanical drawing	20
SICIP-LE-CAD-02-O	Perform CAD modeling	1. Install CAD software 2. Create 2D drawing 3. Develop 3D model 4. Create mold based on 3D model	100
SICIP-LE-CAD-03-O	Select cutting tools and workpiece	1. Understand cutting tools 2. Select cutting tools for lathe operation 3. Select cutting tools for milling operation	20
SICIP-LE-CAD-04-O	Perform CAM programming	1. Install CAM Software 2. Use CAM software interface 3. Carry out 2D CAM programming 4. Carry out 3D CAM programming 5. Load and run program	110
SICIP-LE-CAD-05-O	Carry out CNC milling machine operations	1. Prepare CNC milling machine 2. Download and transfer program 3. Operate CNC milling machine 4. Check and measure work piece	20
SICIP-LE-CAD-06-O	Carry out CNC lathe machine operations	1. Prepare CNC lathe machine 2. Download and transfer program 3. operate CNC lathe machine 4. Check and measure work piece	20
Total Hour			290

COMPETENCY STANDARD: CAD-CAM DESIGN AND PROGRAMMING (MECHANICAL)

A. The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-CAD-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards.

	3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures. 1.4 Reported incidents arising from hazards and risks to authority. 1.5 Implemented plans and procedures to respond emergency. 1.6 Applied basic first aid procedure.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 05 hrs.	Unit Code: SICIP-LE-CAD-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading and understanding workplace documents in English, preparing simple workplace documents in English, listening and comprehending to English conversations and performing conversations in English.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read and understand workplace documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted.
2. Prepare simple workplace documents in English	2.1 Simple <u>routine workplace documents</u> are prepared using key words, phrases, simple sentences 2.2 <u>Visual aids</u> are prepared and submitted. 2.3 Key information is written in the appropriate places in standard forms.
3. Listen and comprehend to English conversations	3.1 Active listening is demonstrated.
4. Perform conversations in English	4.1 Conversation is performed in English with peers, customers and management as per the workplace standard.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Routine workplace documents	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs 2.6 Charts

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Interpret workplace documents in English 1.2 Write simple routine workplace documents in English 1.3 Listen to conversation in English 1.4 Perform conversation in English 1.5 Interaction skills (i.e., teamwork, interpersonal skills, etc.) 1.6 Job roles, responsibilities and compliances
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2. Underpinning Skills	2.1 Ability to read and understand workplace documents in English by using appropriate vocabulary and grammar, standard spelling and punctuation 2.2 Ability to write simple routine workplace documents in English such as: Schedules and agenda, job sheets, operational manuals and brochures and promotional material 2.3 Ability of listening in English and interpreting 2.4 Ability to perform conversation in English with peers, customers and management 2.5 Work effectively with others 2.5.1 Listening and questioning skills 2.5.2 Ability to follow simple directions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, material and documentation required 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Conversed in English with peers and customers. 1.2 Made reports of workplace documents in English.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 05 hrs.	Unit Code: SICIP-LE-CAD-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued 2.9 Identifying current and potential problems faced by the team 2.10 Identifying solutions to the problem 2.11 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

B. The Sector Specific (Common) Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 08 hrs.	Unit Code: SICIP-LE-CAD-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of wastes</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce wastes in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage

	2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

2. Underpinning Knowledge	2.1 Principles of green practices in Light Engineering sector 2.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 2.3 Sources of environmental impacts in Light Engineering 2.4 Methods to minimize resource consumption (water, energy, raw materials) 2.5 Waste management practices 2.6 Sustainable alternatives to fossil-based energy resources 2.7 Personal and workplace habits for reducing environmental impact
4. Underpinning Skills	4.1 Identifying environmental impacts in Light Engineering activities 4.2 Applying methods to minimize resource use in the workplace 4.3 Implementing waste management practices 4.4 Following procedures to safely dispose of hazardous materials

	4.5 Documenting water, energy, and material consumption in the workplace 4.6 Performing green practices in personal and professional activities
5. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
7. Resource Implications	7.1 Workplace (simulated or actual) 7.2 Different types of Light Engineering hand tools and power tools 7.3 Safety gear 7.4 Pens 7.5 Papers 7.6 Work books 7.7 Operation and maintenance manuals 7.8 Waste segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Light Engineering 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK WITH HAND TOOLS AND POWER TOOLS	Nominal Duration: 08 hrs.	Unit Code: SICIP-LE-CAD-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools and power tools for usability, using hand tools properly and safely, operating power tools properly and safely, and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools and power tools for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools</u> and <u>power tools</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools properly and safely	2.1 Appropriate hand tool for the job is used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> are observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing.
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance with manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 <u>Measuring tools</u> are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Hand tools	1.1 Screwdrivers 1.2 Pliers 1.3 Wire Strippers 1.4 Fish Wire (for pulling wire through conduit) 1.5 Leveler 1.6 Tester 1.7 Allen key 1.8 Nut Drivers 1.9 Crimping Tool (for wire terminals)

	1.10 Flashlight or Headlamp 1.11 Hacksaw (for cutting conduit) 1.12 Chisels and Hammer 1.13 Adjustable wrench 1.14 File 1.15 Socket wrench 1.16 Ratchet
2. Power tools	2.1 Power drills 2.2 Nail guns 2.3 Angle grinders and Grinders 2.4 Pneumatic wrenches 2.5 Pedestal drills
3. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition of hand tools and power tools 3.5 Awareness to OHS requirements
4. Measuring tools	4.1 Tachometer 4.2 Vernier Calipers 4.3 Calipers 4.4 Micrometer 4.5 Bevel Protractor

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and techniques 1.3 Types of Power tools, use and safe handling method 1.4 Technical application of tools 1.5 Procedures in the use of hand tools and power tools 1.6 Policies and procedures for occupational health and safety 1.7 Use of PPE 1.8 Handling of tools and equipment 1.9 Reporting and documentation 1.10 Preventive maintenance 1.11 Methods and techniques 1.12 Quality procedures 1.13 Waste management 1.14 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tool for the job 2.2 Observing safety precautions when using hand tools 2.3 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.4 Checking condition of tools after use 2.5 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.6 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.7 Storing Tools and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-CAD-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry-out measurements and calculations. It specifically includes the tasks of checking usability of measuring devices, carrying out accurate work measurements, executing simple work calculations, cleaning and maintaining measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate <u>measuring device</u> is selected for the job. 1.2 Application of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate work measurements	2.1 Measurements are obtained using appropriate measuring devices. 2.2 <u>Systems of measurements</u> are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple work calculations	3.1 Simple calculations involving <u>four basic mathematical calculations</u> are executed. 3.2 Other operations are used to complete tasks in Light Engineering works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign materials are removed from measuring instrument. 4.2 Condition of instrument is checked. 4.3 Appropriate lubricant is applied after use and prior to storage. 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance with workplace procedure.

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Slide calipers 1.2 Steel tape measure 1.3 Steel rule 1.4 Square 1.5 Tri-square 1.6 Feeler gauges 1.7 Water level 1.8 Hose level 1.9 Thermometers 1.10 Protractor
2. Systems of measurements	2.1 Metric system 2.2 FPS system

3. Four basic mathematical Calculations	3.1 Area Calculation 3.2 Volume Calculation 3.3 Weight Calculation 3.4 Angle Calculation
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of measuring and checking tools/instruments 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Measuring tools' operating and maintenance manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test

	2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

C. Occupation-specific Competencies

Unit of Competency: UNDERSTAND MECHANICAL DRAWING	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-CAD-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand mechanical drawing. It specifically includes the tasks of interpreting mechanical drawing and creating mechanical drawing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret mechanical drawing	1.1 Different types of <u>mechanical drawings</u> are interpreted. 1.2 Different <u>views of mechanical drawing</u> are identified and described. 1.3 Necessity of putting dimensions in different drawings is comprehended.
2. Create mechanical drawing	2.1 Drawing is created accurately. 2.2 Components, objects, parts, materials and assemblies are identified, drawn and described. 2.3 Base line or datum points are specified, as required. 2.4 Instructions are included in drawing. 2.5 <u>Tolerance</u> , limits and fits are identified in technical drawing.

Range of Variables

Variable	Range
	May include but not limited to:
1. Mechanical drawings	1.1 Sketch components 1.2 Line 1.3 Rectangle 1.4 Circle 1.5 Ellipse 1.6 Polygon 1.7 Arc 1.2 Elements to determine dimension 1.2.1 Central line 1.2.2 Object line 1.2.3 Tolerance 1.2.4 Ladder line 1.2.5 Extension line 1.2.6 Hidden line 1.2.7 Cutting plane line 1.2.8 Short brake 1.2.9 Long brake
2. Views of mechanical drawing	2.1 Top and Bottom View 2.2 Right and Left View 2.3 Front View 2.4 Sectional view 2.5 Isometric View

	2.6 Oblique View 2.7 Hidden view 2.8 Perspective view
3. Tolerance	3.1 General 3.2 Angular 3.3 Geometric

Curricular Content Guide

1. Underpinning knowledge	1.1 Drawing tools and supplies 1.2 Drawing symbols 1.3 Different views of drawing 1.4 Dimensioning techniques 1.5 Tolerance, limits and fits 1.6 Engineering materials
2. Underpinning skills	2.1 Interpreting mechanical drawing 2.2 Creating mechanical drawing
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Drawing tools and supplies 4.4 Specimen parts and components 4.5 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Interpreted mechanical drawing 1.2 Created mechanical drawing
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM CAD MODELING	Nominal Duration: 100 hrs.	Unit Code: SICIP-LE-CAD-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform CAD modelling. It specifically includes the tasks of installing CAD software, creating 2D drawing, developing 3D model and creating mold based on 3D model.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install CAD software	1.1 <u>Components of computer</u> are identified and interpreted. 1.2 CAD software is installed as per standard operating procedure. 1.3 Screen display areas and <u>basic parameters</u> are set as per job specification.
2. Create 2D drawing	2.1 <u>Sketch tools</u> are identified and selected for 2D drawing. 2.2 <u>Sketch modify tools</u> are identified and selected for 2D drawing. 2.3 2D <u>sketch relations</u> are identified and explained. 2.4 2D drawing is created. 2.5 CAD drawing is reviewed and modified, as necessary.
3. Develop 3D model	3.1 <u>Features tools</u> are identified and selected for 3D model. 3.2 <u>Direct editing tools</u> are identified and selected for 3D model. 3.3 3D model is developed.
4. Create mold based on 3D model	4.1 3D model is reviewed and modified, if necessary. 4.2 2D drawing is generated for review and finalizing the model. 4.3 Mold is created based on the 3D model. 4.4 Mold is converted to <u>CAM supported files.</u> 4.5 A separate folder is developed and the mold is saved for CAM programming.

Range of Variables

Variable	Range
	May include but not limited to:
1. Components of computer	1.1 Hardware 1.1.1 RAM 1.1.2 Processor 1.1.3 Graphic Card 1.1.4 Storage 1.1.5 Monitor 1.2 Operating system 1.3 Software 1.3.1 Solid works 1.3.2 Siemens NX 1.3.3 CATIA 1.3.4 Auto CAD 1.3.5 Rhino 1.3.6 Fusion 360
2. Basic parameters	2.1 User interface

	2.2 Units (mm or inch) 2.3 Color and text format
3. Sketch tools	3.1 Lines 3.2 Rectangle 3.3 Arc 3.4 Circle 3.5 Polygon 3.6 Ellipse 3.7 Spline 3.8 Slot
4. Sketch modify tools	4.1 Trim 4.2 Offset 4.3 Mirror 4.4 Linear and circular pattern 4.5 Move, copy, rotate, scale, extend and stretch
5. Sketch relations	5.1 Horizontal 5.2 Vertical 5.3 Coincident 5.4 Intersection 5.5 Tangent 5.6 Collinear 5.7 Parallel 5.8 Concentric 5.9 Pairs 5.10 Symmetric
6. Feature tools	6.1 Extruded boss/base 6.2 Extruded cut 6.3 Revolved boss/base 6.4 Revolved cut 6.5 Swept boss/base 6.6 Swept cut 6.7 Lofted boss/base 6.8 Lofted cut 6.9 Boundary boss/base 6.10 Boundary cut
7. Direct editing tools	7.1 Fillet, Chamfer, Rib, Draft, Shell, Wrap, Mirror, Pattern 7.2 Flex 7.3 Surface tools 7.4 Mold tools 7.5 Move face 7.6 Move/copy bodies 7.7 Delete face 7.8 Split 7.9 Combine
8. CAM supported files	8.1 Step 8.2 iges 8.3 Parasolid

Curricular Content Guide

1. Underpinning knowledge	1.1 Components of computer 1.2 CAD software 1.3 2D drawings 1.4 3D model 1.5 Mold
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2. Underpinning skills	2.1 Operating computer 2.2 Installing and using CAD software 2.3 Creating 2D drawing 2.4 Developing 3D model 2.5 Creating mold 2.6 Converting mold into CAM supported files
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Stationery 4.4 Learning manual 4.5 Software 4.6 Printer/Plotter 4.7 Tools and attachments 4.8 Drawings, Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Identified and interpreted job requirements 1.2 Installed CAD software 1.3 Created 2D drawing 1.4 Developed 3D model 1.5 Created mold based on 3D model 1.6 Converted mold into CAM supported files
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: SELECT CUTTING TOOLS AND WORKPIECE	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-CAD-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to select cutting tools and workpiece. It specifically includes the tasks of understanding cutting tools, selecting cutting tools for lathe operation and selecting cutting tools for milling operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand cutting tools	1.1 <u>Geometry of cutting tools</u> is identified. 1.2 <u>Types of cutting tools</u> are identified. 1.3 <u>Cutting parameters</u> are identified.
2. Select cutting tools for lathe operation	2.1 <u>Cutting tools for lathe operation</u> are identified and the use of tools is described. 2.2 <u>Workpiece material</u> is identified. 2.3 Cutting tools for lathe operation are selected for use.
3. Select cutting tools for milling operation	3.1 <u>Cutting tools for milling operation</u> are identified and the use of tools is described. 3.2 Cutting tools for milling operation are selected for use.

Range of Variables

Variable	Range
	May include but not limited to:
1. Geometry of cutting tools	1.1 Number of flutes 1.2 Diameter and length 1.3 Type of insert 1.3.1 CNMG 1.3.2 TNMG 1.3.3 DNMG 1.3.4 VNMG 1.4 Material of cutting tool 1.4.1 High speed steel 1.4.2 Carbide 1.5 Angles of cutting tool 1.5.1 Rack angle 1.5.2 Clearance angle 1.5.3 Relief angle
2. Types of cutting tools	2.1 Single point 2.1.1 Turning 2.1.2 Facing 2.1.3 Boring 2.1.4 Threading 2.2 Multi point 2.2.1 Solid end mill 2.2.2 Ball mill 2.2.3 Radius mill 2.2.4 Face mill 2.2.5 Drill bit

	2.2.6 Reamer 2.2.7 Tap and die
3. Cutting parameters	3.1 Cutting speed 3.2 Feed 3.3 Depth of cut 3.4 RPM
4. Cutting tools for lathe operation	4.1 Turning tools 4.2 Facing tools 4.3 Grooving tools 4.4 Threading tools 4.5 Boring tools
5. Workpiece material	5.1 Aluminum 5.2 Mild steel 5.3 Teflon
6. Cutting tools for milling operation	6.1 Face mill 6.2 End mill 6.3 Bull mill 6.4 Ball mill 6.5 Machine tap

Curricular Content Guide

1. Underpinning knowledge	1.1 Job requirements and workpiece specifications 1.2 Cutting tools, equipment and materials (for lathe and milling machines) 1.3 Setting of cutting tools 1.4 Cutting parameters (speed, feed, depth of cut, RPM)
2. Underpinning skills	2.1 Establishing job requirements and work piece specifications 2.2 Identifying and selecting correct tools and attachments 2.3 Setting-up work to using appropriate tools and clamping device 2.4 Setting work to avoid distortion on release of clamping devices 2.5 Selecting, calculating and using appropriate speed, feed and depth of cut 2.6 Planning and sequencing of operations 2.7 Checking conformance to specifications 2.8 Performing any necessary corrective measures to workpiece(s)
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Materials 4.4 Stationery 4.5 Learning manual

	4.6 Software 4.7 Tools and attachments 4.8 Drawings, specifications and work instructions 4.9 Learning manual
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Set-up workpiece using appropriate tools and clamping devices 1.2 Selected, calculated and used appropriate speed, feed rate and depth of cut 1.3 Checked and measured work piece against specifications 1.4 Performed any necessary corrective measures to work piece
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM CAM PROGRAMMING	Nominal Duration: 110 hrs.	Unit Code: SICIP-LE-CAD-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform CAM programming. It specifically includes the tasks of installing CAM software, using CAM software interface, carrying out 2D CAM programming, carrying out 3D CAM programming, loading and running program.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install CAM Software	1.1 Materials, instructions and standard operating procedure are obtained according to job requirement. 1.2 CAM package is installed as per standard operating procedure. 1.3 <u>System parameters</u> are identified and selected according to job requirement.
2. Use CAM software interface	2.1 <u>Software interface components</u> are identified and described. 2.2 CAM software interface is used as per job requirements.
3. Carry out 2D CAM programming	3.1 CAD model is imported and oriented. 3.2 Reference point is established or originated based on job requirement. 3.3 Stock set-up is performed. 3.4 Cutting tools are identified and selected. 3.5 Sequential <u>2D toolpaths for milling operations</u> are identified, generated and verified. 3.6 Sequential <u>2D toolpaths for lathe operations</u> are identified, generated and verified. 3.7 <u>G code</u> and <u>M code</u> are generated.
4. Carry out 3D CAM programming	4.1 3D CAD model is imported and oriented. 4.2 Reference point is established or originated based on job requirement. 4.3 Stock set-up is performed. 4.4 Cutting tools are identified and selected 4.5 Sequential <u>3D toolpaths for milling operations</u> are identified, generated and verified. 4.6 G code and M code are generated.
5. Load and run program	5.1 Program is loaded using appropriate <u>device</u> . 5.2 Dry run/simulation is performed in computer and machine as per standard operating procedure. 5.3 Program is executed to produce work piece. 5.4 <u>Problems</u> encountered are recorded and reported to appropriate authority as per standard operating procedure.

Range of Variables

Variable	Range
	May include but not limited to:
1. System parameters	1.1 Metric or English system 1.2 Tool bars (dimensioning, line types, editing, hatching)
2. Software interface	2.1 Wireframe

components	2.1.1 Line 2.1.2 Rectangle 2.1.3 Circle 2.1.4 Polygon 2.1.5 Slot 2.1.6 Arc 2.1.7 Rectangular shape 2.1.8 Carve on one edge 2.1.9 Bounding box 2.1.10 Turn profile 2.2 Surface 2.2.1 Move face 2.2.2 Delete face 2.2.3 Push/pull 2.2.4 Modify 2.3 Solid 2.3.1 Extrude boss/cut 2.3.2 Revolve boss/cut 2.3.3 Lofted boss/cut 2.4 Transform 2.4.1 Translate 2D 2.4.2 Translate 3D 2.4.3 Dynamic transform 2.4.4 Move to origin 2.4.5 Offset 2.4.6 Mirror 2.4.7 Pattern 2.5 Machine 2.5.1 CNC default lathe machine 2.5.2 CNC default milling machine 2.6 Toolpath 2.6.1 Facing 2.6.2 Pocket milling 2.6.3 Contour 2.6.4 Parallel 2.6.5 Scallop 2.6.6 Drilling
3. 2D toolpaths for milling operations	3.1 Facing 3.2 Pocket milling 3.3 Contour 3.4 Drilling
4. 2D toolpaths for lathe operations	4.1 Roughing 4.1.1 Turning 4.1.2 Boring 4.2 Finishing 4.2.1 Turning 4.2.2 Boring 4.3 Drilling 4.3.1 Direct 4.3.2 Peck 4.4 Parting 4.5 Threading 4.5.1 External 4.5.2 Internal

	4.6 Grooving
5. G code	5.1 G00 5.2 G01 5.3 G02 5.4 G03 5.5 G17, G18, G19 5.6 G20, G21 5.7 G28 5.8 G41, G42, 5.9 G43, G49 5.10 G54-G59 5.11 G90, G91
6. M code	6.1 M01, M02, M03, M04 6.2 M05, M06 6.3 M08, M09 6.4 M30
7. 3D toolpaths for milling operations	7.1 Roughing 7.1.1 Surface rough pocket 7.1.2 Surface rough parallel 7.2 Finishing 7.2.1 Surface finish contour 7.2.2 Surface finish scallop
8. Device	8.1 USB drive 8.2 Memory card 8.3 Portable hard disk 8.4 Networking
9. Problems	9.1 Incorrect machine set-up 9.2 Incorrect parameter setting 9.3 Incorrect tool and work offset 9.4 Defective raw materials

Curricular Content Guide

1. Underpinning knowledge	1.1 CAM software 1.2 System parameters 1.3 Software interface components 1.4 2D CAM programming 1.5 Sequential 2D toolpaths for milling operations 1.6 Sequential 2D toolpaths for lathe operations 1.7 G code and M code 1.8 Sequential 3D toolpaths for milling operations 1.9 Appropriate devices
2. Underpinning skills	2.1 Installing CAM software 2.2 Setting system parameters 2.3 Using CAM software interface 2.4 Carrying out 2D CAM programming 2.5 Carrying out 3D CAM programming 2.6 Applying G code and M code 2.7 Loading and running program using appropriate device 2.8 Recording and reporting problems encountered to appropriate authority
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Stationery 4.4 Learning manual 4.5 Software 4.6 Printer/Plotter 4.7 Tools and attachments 4.8 Drawings, Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Installed CAM Software 1.2 Used CAM software interface 1.3 Carried out 2D CAM programming 1.4 Carried out 3D CAM programming 1.5 Set system parameters 1.6 Applied G code and M code 1.7 Loaded and run program using appropriate device
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT CNC MILLING MACHINE OPERATIONS	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-CAD-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to carry out CNC milling machine operations. It specifically includes the tasks of preparing CNC milling machine, downloading and transferring program, operating CNC milling machine, and checking and measuring work piece.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC milling machine	1.1 Main switch and emergency switch are turned on and voltage stabilizer are checked. 1.2 Lubricant, air & hydraulic pressure and coolant are checked as per manufacturer's specification. 1.3 Machine zero point is checked. 1.4 <u>CNC machine controller</u> interface is identified and described. 1.5 <u>Cutting tools</u> are set according to required sequence of operation. 1.6 <u>Clamping devices</u> are set and tightened according to standard operating procedures. 1.7 Work piece is mounted and centered on clamping device to required level of accuracy as per workplace procedures. 1.8 Part zero setting is performed. 1.9 Tool zero setting is performed for each tool.
2. Download and transfer program	2.1 Program is downloaded and transferred to the machine using appropriate device. 2.2 Program is simulated to determine the correctness of the toolpath and other work parameters.
3. Operate CNC Milling Machine	3.1 <u>CNC milling operations</u> are performed to produce component as per program. 3.2 Corrective measures are performed, if necessary.
4. Check and measure work piece	4.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 4.2 Defective work pieces are marked, recorded and reported for proper action.

Range of Variables

Variable	Range
	May include but not limited to:
1. CNC machine controller	1.1 Haas 1.2 Fanuc 1.3 Siemens 1.4 Mitsubishi 1.5 GSK
2. Cutting tools	2.1 Face/Bull Mill 2.2 End Mill 2.3 Ball mill 2.4 Tapper mill 2.5 Drill Bits 2.6 Reamers

3. Clamping devices	3.1 Angle plate 3.2 V-Block, U-Clamp and C-Clamp 3.3 Step-Block 3.4 T-Slot bolt 3.5 Machine vice 3.6 Pneumatic fastening clamps 3.7 Jig and fixtures
4. CNC milling operations	4.1 Facing 4.2 Pocket 4.3 Drilling 4.4 Contour 4.5 3D toolpaths

Curricular Content Guide

1. Underpinning knowledge	1.1 Job requirements and work piece specifications 1.2 Tools, equipment and materials 1.3 Set-up of CNC milling machine 1.4 Setting of cutting tools 1.5 Procedure for checking lubricant and coolant 1.6 Procedure for checking air and hydraulic pressure 1.7 Procedure for setting machine zero point 1.8 CNC milling machine operations 1.9 Measuring tools and corrective measures
2. Underpinning skills	2.1 Establishing job requirements and work piece specifications 2.2 Identifying and selecting correct tools and attachments 2.3 Setting-up work to using appropriate tools and clamping device 2.4 Setting work to avoid distortion on release of clamping devices 2.5 Selecting, calculating and using appropriate speed, feed and depth of cut 2.6 Planning and sequencing of operations 2.7 Checking conformance to specifications 2.8 Performing CNC milling machine operations using precision measuring equipment 2.9 Checking and measuring work piece against specifications within specified tolerances 2.10 Performing any necessary corrective measures
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Materials 4.4 Stationery 4.5 Learning manual

	4.6 Software 4.7 Personal protective equipment (PPE) 4.8 Milling Tools and attachments 4.9 CNC milling machine 4.10 Drawings, specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Prepared CNC milling machine 1.2 Set-up workpiece using appropriate tools and clamping devices 1.3 Selected, calculated and used appropriate speed, feed rate and depth of cut 1.4 Performed CNC milling machine operations 1.5 Checked and measured work piece against specifications 1.6 Performed any necessary corrective measures to work piece
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT CNC LATHE MACHINE OPERATIONS	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-CAD-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to carry out CNC lathe machine operations. It specifically includes the tasks of preparing CNC lathe machine, downloading and transferring program, operating CNC lathe machine, and checking and measuring work piece.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC lathe machine	1.1 Main switch and emergency switch are turned on and voltage stabilizer are checked. 1.2 Lubricant, air & hydraulic pressure and coolant are checked as per manufacturer's specification. 1.3 Machine zero point is checked. 1.4 CNC machine controller interface is identified and described. 1.5 <u>Cutting tools</u> are set according to required sequence of operation. 1.6 <u>Clamping devices</u> are set and tightened according to standard operating procedures. 1.7 Work piece is mounted and centered on clamping device to required level of accuracy as per workplace procedures. 1.8 Part zero setting is performed for each tool.
2. Download and transfer program	2.1 Program is downloaded and transferred to the machine using appropriate device. 2.2 Program is simulated to determine the correctness of the tool path and other work parameters.
3. Operate CNC lathe machine	3.1 <u>CNC lathe operations</u> are performed to produce component as per program. 3.2 <u>Corrective measures</u> are performed, if necessary.
4. Check and measure work piece	4.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 4.2 Defective work pieces are marked, recorded and reported for proper action.

Range of Variables

Variable	Range
	May include but not limited to:
1. Cutting tools	1.1 Facing 1.2 Turning 1.3 Grooving 1.4 Drilling 1.5 Threading 1.6 Parting-off 1.7 Taping 1.8 Boring 1.9 Knurling

	1.10 Honing 1.11 Finishing
2. Clamping devices	2.1 Three jaw chuck 2.2 Four jaw chuck 2.3 Collet chuck 2.4 Live center 2.5 Bar feeder 2.6 Part catcher 2.7 Tool center
3. CNC lathe operations	3.1 Facing 3.2 Turning 3.3 Tapper Turning 3.4 Threading 3.5 Drilling 3.6 Boring 3.7 Taping 3.8 Parting-off
4. Corrective measures	4.1 Cutting tool changes 4.2 Adjustment of tool offset 4.3 Adjustment of cutting speed and feed rate

Curricular Content Guide

1. Underpinning knowledge	1.1 Job requirements and work piece specifications 1.2 Tools, equipment and materials 1.3 Set-up of CNC lathe machine 1.4 Setting of cutting tools 1.5 Procedure for checking lubricant and coolant 1.6 Procedure for checking air and hydraulic pressure 1.7 Procedure for setting machine zero point 1.8 CNC lathe machine operations 1.9 Measuring tools and corrective measures
2. Underpinning skills	2.1 Establishing job requirements and work piece specifications 2.2 Identifying and selecting correct tools and attachments 2.3 Setting-up work to using appropriate tools and clamping device 2.4 Setting work to avoid distortion on release of clamping devices 2.5 Selecting, calculating and using appropriate speed, feed and depth of cut 2.6 Planning and sequencing of operations 2.7 Checking conformance to specifications 2.8 Performing CNC lathe machine operations using precision measuring equipment 2.9 Checking and measuring work piece against specifications within specified tolerances 2.10 Performing any necessary corrective measures
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Personal protective equipment (PPE) 4.4 Materials 4.5 Stationery 4.6 Learning manual 4.7 Software 4.8 Lathe Tools and attachments 4.9 CNC lathe machine 4.10 Drawings, specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 3.1 Prepared CNC lathe machine 3.2 Set-up workpiece using appropriate tools and clamping devices 3.3 Selected, calculated and used appropriate speed, feed rate and depth of cut 3.4 Performed CNC lathe machine operations 3.5 Checked and measured work piece against specifications 3.6 Performed any necessary corrective measures to work piece
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	CAD-CAM Design and Programming (Mechanical)
Number of Trainees:	25

Course-wise Training Space (Computer Lab and CNC Workshop):

- Computer Lab : 400 sft
- CNC Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Workstation computer with CAD and CAM software	26
2.	CNC Lathe Machine	1
3.	CNC Milling Machine	1
4.	Bench Vice	5
5.	Working Table	2
6.	Cutting tools set for lathe machine	2
7.	Cutting tools set for milling machine	2
8.	Vernier caliper	5
9.	Micrometer	5

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on CAD-CAM Design and Programming (Mechanical), the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.

CAD-CAM Design and Programming (Mechanical)